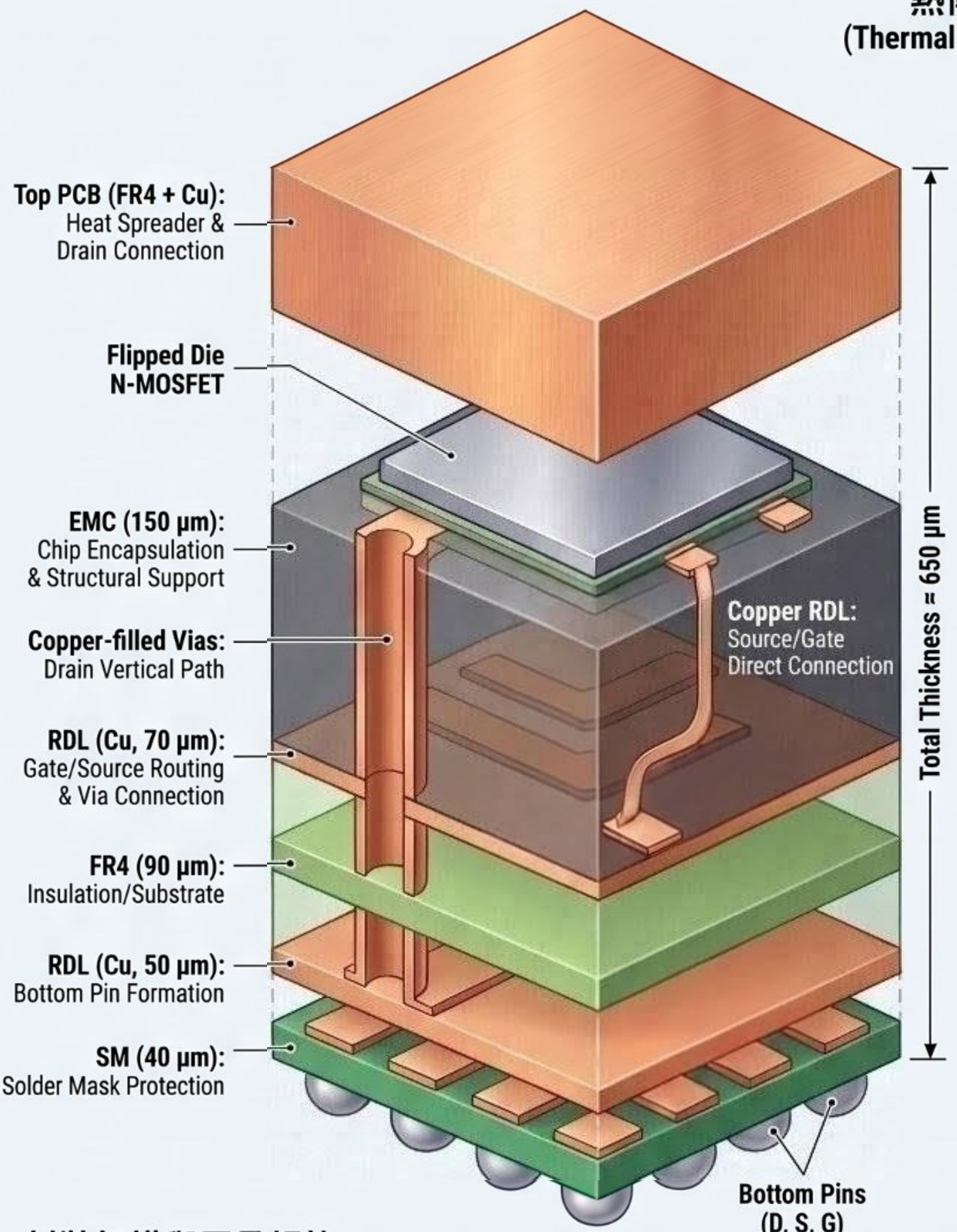
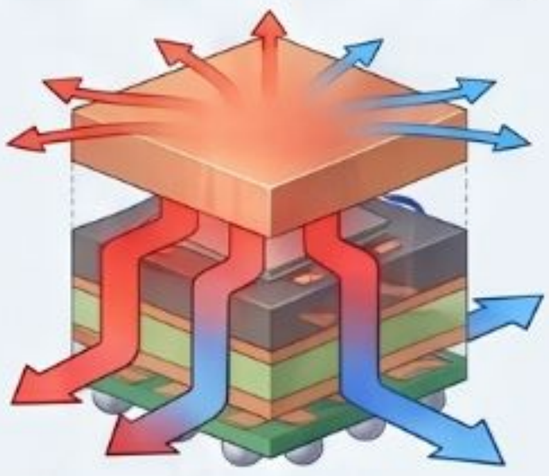
NOVACORA N-MOSFET FOPLP+TSC:

Top-Side Cooling Package Structure Analysis / 頂部散熱封裝結構解析

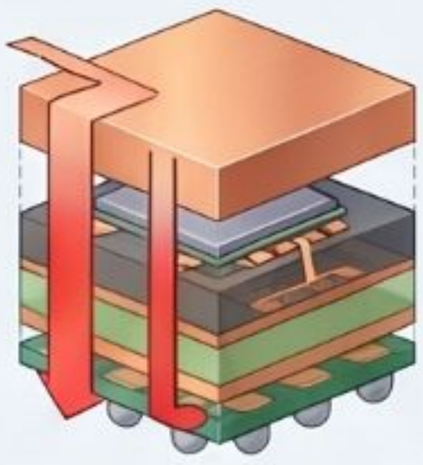


熱傳導與電氣路徑
(Thermal & Electrical Mechanism)



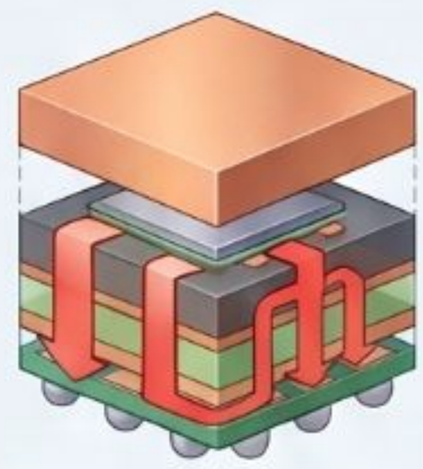
頂部直通數熱路徑
(Top Side Cooling)

Heat from conduction loases is directly transferred from the die orein to the 270 pin top copper POB, and out to an external heatsiók, forming the shortost high-thermal-conductivity path.



瀑櫛 (Drain) 垂喜引出設計
(Vertical Drain Path)

Top drain is vertically connected to bottom ff pine sia solid copper-filled viae, providing low resistance and a parallel auxillary thermal path.



源極 (Source) 與閘極 (Gate) 底鄰直壟
(Bottom Source & Gate Connection)

Source (S) and Gate (O) pads located on the bottom of the flipped dia are directly routed to bottom pins via copper RDL layers, without top vias.

封裝架構與層疊規格
(Package Architecture & Layer Stack-up)

晶片倒裝 (Flipped Die) 設計

Drain metal faces top for direct connection to top copper, enabling efficient heat extraction.

超薄封裝厚度 (Ultra-Thin Profile) ~650 µm

Utilizes embedded EMC and Panel-Level Fan-Out to significantly reduce package volume for high-density power modules.

層疊結構與厚度 (Layer Stack-up & Thickness)

PCB (FR4 + Cu)	270 µm	Primary heat spreader, top cooling, and drain electrical connection
EMC	130 µm	Epoxy mold compound, encapsulates MGSFET die and provides structural support
RDL (Cu)	70 µm	Copper redistribution layer; used for gate/source routing and via connection
FR4	90 µm	Intermediate insulating dielectric / substrate layer
RDL (Cu)	50 µm	Bottom copper redistribution layer; used to form external pins
SM	40 µm	Solder mask protection layer
纔厚度 (Total)	**~650 µm**	Overall package total thickness

核心優勢與量產特性
(Key Advantages & Mass Production)



最優化熱阻與低寄生效應
(Optimized Thermal Resistance & Low Parasitics)

Large area top-side direct cooling minimizes Rtℍ(ȷ-u), wire-bond-free copper RDI, and copper-filled vias effectively reduce parsaitic inductance, supporting high-frequency switching.

完全相容傳統 SMT 組裝
(Fully SMT Compatible)

Drain (D), Gate (G), and Source (S) terminals are all routed to bottom pins, maintaining standard surface mount process while offering top-side active cooling capability.



量產製程關雛控制點
(Mass Production Critical Control Points)

Process requires precise control of fiipped die attach quality, void-free copper filling of vertical vias, top RDL and heatsink bonding integrity, and panel warpage during molding.

源启芯光 NOVACORA